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Fracture Stimulated Volume Calculation Method for Horizontal Wells

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Abstract

This paper aims to present and validate an efficient numerical method based on the Fast Marching Method (FMM) for accurately calculating the fracture-controlled stimulated reservoir volume (SRV) in horizontal wells, particularly within heterogeneous reservoirs. The methodology focuses on rapid evaluation and optimization of hydraulic fracturing design parameters (e.g., cluster spacing, fracture conductivity) and providing theoretical and practical guidance for effective unconventional resource development. The Eikonal equation, derived from pressure diffusion asymptotics to track the pressure front (Diffusive Time of Flight, DTOF), is solved numerically using the efficient Fast Marching Method (FMM). This yields the DTOF field, from which stimulated reservoir volume (SRV) and its derivative are calculated via DTOF-to-physical-time conversion; the derivative aids flow regime identification. Accuracy is confirmed against analytical solutions for homogeneous radial/linear flow. Parametric sensitivity analyses and a field case demonstrate the method's practical utility in optimizing designs and guiding field development. The proposed FMM -based SRV calculation method demonstrates good accuracy and significantly high computational efficiency, showing excellent agreement with analytical solutions during early-time, unbounded flow. The SRV evolution curves and their derivatives effectively identify distinct flow regimes, such as fracture linear flow, inter-fracture interference, and boundary-dominated flow. Permeability is identified as a critical factor governing SRV extent; for micro/nano-Darcy reservoirs, the pressure propagation distance is extremely limited, highlighting the paramount importance of smaller cluster spacing for effective stimulation. Increasing fracture intensity (i.e., reducing cluster spacing) significantly enhances SRV and reserve recovery. A key finding is that high-density fracturing strategies can effectively reduce the reservoir's dependency on high fracture conductivity. This theoretical prediction is strongly corroborated by a field pilot in the Chang 7 shale oil play, Ordos Basin: optimizing cluster spacing from 12m to 5-8m resulted in a 31.3% increase in average cumulative oil production per well over two years for the test group, confirming the significant EOR effect of high-density fracturing. This work innovatively applies the Fast Marching Method to the dynamic assessment of SRV in multi-fractured horizontal wells, offering a practical tool that balances computational efficiency and accuracy, especially for heterogeneous reservoirs. The study not only theoretically reveals the underlying mechanism by which high-density fracturing reduces

the demand for high fracture conductivity but also validates its practical value in enhancing unconventional resource recovery through field application data, providing new perspectives and robust technical support for optimizing fracturing designs and improving recovery factors in the industry.

Introduction

A critical challenge in the development of unconventional resources is the accurate and efficient evaluation of the stimulated reservoir volume (SRV)(Mayerhofer, 2008; Cipolla, 2010), which directly informs the optimization of completion strategies such as cluster and well spacing (Yu and Rahman, 2012). The application of pressure diffusion theory, particularly through the Diffusive Time-of-Flight (DTOF) concept solved by the Fast Marching Method (FMM), has emerged as a cornerstone for rapid reservoir analysis. This paper builds upon this foundation to present a fast and robust method for calculating and optimizing the fracture-stimulated volume. Foundational research in this domain demonstrated the power of using DTOF to link complex reservoir models with dynamic data. For instance, Li and King (2020) established a novel semi-analytical method that utilizes FMM to solve the Eikonal equation. By transforming the complex 3D pressure diffusion problem into an equivalent 1D problem along the DTOF coordinate, their work showed it was possible to efficiently integrate pressure transient data to calibrate permeability fields in heterogeneous reservoir models, even providing reasonable updates beyond the conventional depth of investigation.Building on this concept, the practical application of FMM has been extended to comprehensive engineering workflows. Paryani et al. (2018) integrated an FMM-based reservoir simulator into a geologically and geomechanically constrained workflow to optimize completion design and well spacing in the Eagle Ford shale. Their study highlighted that the FMM-based simulator could provide extremely fast estimates of drainage area and pressure depletion, thereby overcoming the significant computational bottleneck of traditional reservoir simulation and enabling timely, adaptive field development decisions. Further demonstrating its versatility, the FMM has also been applied to the detailed analysis and simplification of complex fracture systems. Xue et al. (2019) combined FMM with spectral clustering to analyze and simplify massive hydraulic fracture networks (HFN) for simulation with Embedded Discrete Fracture Models (EDFM). In their work, the DTOF calculated by FMM served as a powerful diagnostic tool for simulation-free well ranking and partitioning of the fracture network. This approach achieved a computational speedup of over 16 times with less than 5% error, proving its efficacy for field-scale models with millions of fracture cells. To address the challenge of applying these methods to increasingly complex and realistic reservoir grids, a significant methodological advancement was introduced. Chen et al. (2021) developed a novel finite-volume formulation of the Eikonal equation. This approach decouples the DTOF calculation from specific grid geometries, relying instead on universal simulator inputs like cell volumes and inter-cell transmissibilities. Their work successfully demonstrated that FMM could be robustly applied to arbitrary grid systems, including unstructured PEBI grids and EDFM, making FMM-based diagnostics a truly versatile and powerful tool for modern reservoir engineering.

The precise characterization of fracture-stimulated reservoir volume constitutes a critical technical challenge in multi-stage hydraulic fracturing of horizontal wellbores, which serves as a crucial factor in optimizing cluster spacing and well spacing. This paper introduces a method that utilizes the Fast Marching Method to calculate the fracture stimulated volume. This method offers the advantage of fast calculation and suitability for non-homogeneous formations.

Assessment criteria

The degree of fracture stimulation volume, which reflects the effectiveness of multi-stage fracturing, can be characterized as

$$\eta = \frac{V_{\mathrm{p}}}{G} \tag{1}$$

Here, Vp represents the drainage volume within the stimulated area, while G denotes the geological reserves in that same area. The drainage volume refers to the pressure decline observed over a specific production period.

The calculation of the drained degree of reserve (transformation volume), denoted as Vp, is the fundamental aspect of Eq. (1). This calculation encompasses direct and indirect methods. The direct method involves numerical simulation of the reservoir, which provides comprehensive consideration of the formation factors. However, it involves a large amount of calculation, which restricts its applicability for theoretical analysis and design purposes. On the other hand, the indirect method primarily employs well testing. Currently, transient flow rate or pressure analysis, including analytical and semi-analytical methods, is predominantly utilized to determine the flow pattern within the formation. Due to the extremely low permeability of unconventional formations, the well testing method requires longer monitoring periods, resulting in limited application.

To efficiently assess the stimulated volume corresponding to the cluster spacing in fracturing design, this paper employs the calculation of radius of investigation. The radius of investigation refers to the radius of detection in vertical wells. Analytical solutions exist for formations with simple flow patterns, such as homogeneous and radial flow. However, a simple analytical formula for the radius of investigation in horizontal wells with multi-stage fracturing is currently unavailable.

Vasco et al. (2000) derived a high-frequency asymptotic solution for the pressure diffusion equation, which corresponds to the Eikonal equation governing pressure wave propagation and controls the formation radius of investigation. The Eikonal equation can be efficiently solved using the fast marching method (Sethian, 1999). Datta-Gupta et al. (2011) conducted an extensive series of studies on this equation and its generalization. In this paper, we utilize the Eikonal equation, which describes pressure wave diffusion, to determine the stimulated volume. We solve the equation using the fast marching method in a 5-point format. Section 2 outlines the method's principle, Section 3 presents an arithmetic comparison to validate the accuracy of the procedure, and Section 4 demonstrates a parametric analysis.

Numerical model

Governing Equation for Pressure Front Propagation

The pressure diffusion equation for a non-homogeneous reservoir is

$$\phi(\boldsymbol{x})\mu c_t(\boldsymbol{x})\frac{\partial p(\boldsymbol{x},t)}{\partial t} = \nabla\cdot[k(\boldsymbol{x})\nabla p(\boldsymbol{x},t)] \tag{2}$$

where: k(x), ϕ(x), ct(x), etc. are functions of location.

The pressure p is Fourier transformed, so that equation (2) has the form in frequency domain space:

$$\phi(\boldsymbol{x})\mu c_t(\boldsymbol{x})(i\omega)p(\boldsymbol{x},\omega) = k(\boldsymbol{x})\nabla^2 p(\boldsymbol{x},\omega) + \nabla k(x)\cdot\nabla p(\boldsymbol{x},\omega) \tag{3}$$

In the frequency domain, the asymptotic solution is expressed as: (Vasco et al. 2000):

$$p(\boldsymbol{x},\omega) = e^{-\sqrt{-i\omega}\,\tau(x)}\sum_{k=0}^{\infty}\frac{A_k(\boldsymbol{x})}{(\sqrt{-i\omega})^k} \tag{4}$$

where: τ(x) is the diffusion time (non-physical time) of the pressure front, representing the propagation time in a non-physical sense, and Ak(x) is the amplitude of the k-th order component of the pressure wave.

In the context of Equation (4), the high-frequency components of the pressure wave propagate the fastest. Therefore, the overall pressure front is determined by the arrival of the highest-frequency component. Under the high-frequency approximation, the series expansion in Equation (4) is dominated by its leading term (k=0). Consequently, we can approximate the full solution by this zeroth-order term:

When k=0, the

$$p(\boldsymbol{x},\omega)=A_0(\boldsymbol{x})e^{-\sqrt{-i\omega}\,\tau(x)} \tag{5}$$

Substituting equation (5) into (3) and finding the highest order term of (3), we obtain the Eikonal equation that governs the pressure front propagation:

$$\sqrt{\alpha(x)}\left|\nabla\tau(x)\right|=1 \tag{6}$$

where a is the diffusion coefficient, defined as

$$\alpha(x)=\frac{k(x)}{\phi(x)\mu c_t(x)} \tag{7}$$

The term $\tau(\mathbf{x})$ represents the diffusive time-of-flight (DTOF). This parameter, which has a physical dimension of T^0.5, should be understood as a non-physical time. Its relationship with the actual physical time, t, is given by:

$$t=\tau^2/c \tag{8}$$

The coefficient c is a constant determined by the flow geometry. Specifically, c=2 for planar linear flow, c=4 for planar radial flow, and c=6 for spherical flow. For more complex geometries or other flow regimes, c must be determined empirically through well test analysis, typically by interpreting pressure-time curves. For the specific case of heterogeneous reservoirs, the methodology for calculating c is detailed in Zhang et al. (2015).

Eikonal equation solving (Fast Marching Method)

The Eikonal equation, as presented in Eq. (6), is a non-linear partial differential equation. It can be efficiently solved using the Fast Marching Method (FMM), a numerical scheme originally proposed by Sethian (1999). The FMM is specifically designed for tracking the propagation of interfaces in boundary value problems. A fundamental requirement of this method is that the speed function governing the front's movement must be single-signed (i.e., either always positive or always negative).

To solve this equation numerically, we first discretize the computational domain using a Cartesian grid with spacings Ax and Ay. By applying a standard five-point finite difference scheme, Equation (6) can be approximated at each grid node (i, j) as follows:

$$\max\left(\frac{\tau_{i,j}-\tau_{i-1,j}}{\Delta x},\frac{\tau_{i,j}-\tau_{i+1,j}}{\Delta x},0\right)+\max\left(\frac{\tau_{i,j}-\tau_{i,j-1}}{\Delta y},\frac{\tau_{i,j}-\tau_{i,j+1}}{\Delta y},0\right)=1\Big/\alpha_{ij} \tag{9}$$

Where: τ_{ij} is the value of the diffusive time-of-flight at the node (x_i,y_j); Δx_s Δy are the dimensions of a grid cell.

The discrete system in Eq. (9) constitutes a finite difference scheme for the Eikonal equation. It adheres to the causality principle through an upwind formulation, and a key advantage of this scheme is its unconditional stability. This system of non-linear equations is solved efficiently using the Fast Marching Method (FMM), which has a computational complexity of $O(N\cdot\log N)$ (N is the number of grids, Sathian 1999). The method is similar to a shortest path algorithm, as each step involves finding the node with the minimum time, which requires the use of a binary heap data structure (Sethian, 1999).

Calculation of the Volume of Investigation, its Derivative, and Bottomhole Pressure

Volume of Investigation. The calculation process begins by imposing a boundary condition at the injection point (i.e., the source node), where its diffusive time-of-flight (DTOF), τ, is initialized to zero. The Fast Marching Method (FMM) then simulates the outward propagation of the pressure front from this source. It systematically computes the DTOF value for every grid node until the entire computational domain is filled with calculated arrival times.

The volume of investigation (VOI), V_p, is given by the following expression:

$$V_p(\tau) = N(\tau)\Delta x \Delta y h \tag{10}$$

where N is the count of grid cells reached by the pressure front at a DTOF of τ, which corresponds to the physical time τ^2/c, and h is the reservoir thickness.

Derivative of the Volume of Investigation. The derivative of the volume of investigation can serve as a characteristic curve for flow regime identification (Xue, 2018; SPE 194017). Therefore, we calculate this derivative in our work. The VOI derivative is given by:

$$w(\tau) = \frac{dV_p(\tau)}{d\tau} \tag{11}$$

Since the raw Vp data is inherently noisy, a smoothing process is required prior to differentiation to avoid generating spurious spikes. In this paper, we apply a time-shifting smoothing algorithm to the data. The method by Xue et al. (2018) was not used because it is more difficult to implement and computationally more expensive.

Bottomhole pressure. Assuming that no flow crosses the boundary of the volume of investigation (i.e., it is treated as a closed, moving boundary), the bottomhole pressure can be obtained from the material balance principle as follows (Xie et al., 2015):

$$\frac{\partial p}{\partial t} \approx \frac{\partial p}{\partial t} = -\frac{1}{c_t}\frac{q_w}{V_p(t)} \tag{12}$$

where q_w is the well production; p is the bottomhole pressure; p is the average pressure; and V_p is the volume of investigation. By integrating Eq. (12), the pressure drawdown, Δp, for a well producing at a constant rate can be directly calculated.

$$\Delta p = \int_0^t \left\{ -\frac{1}{c_t}\frac{q_w}{V_p[r(t)]} \right\} dt \tag{13}$$

Model Verification

Comparison with the Analytical Solution for Radial Flow in a Vertical Well

For the classic case of radial flow towards a vertical well situated in a homogeneous and infinite-acting reservoir, the radius of investigation, r, is calculated as:

$$r = 2\sqrt{\alpha t} \tag{14}$$

The volume of investigation (V_p) for the radial flow case in a homogeneous reservoir is given by the following expression:

$$V_p = \pi r^2 h = 4\pi h \alpha t \tag{15}$$

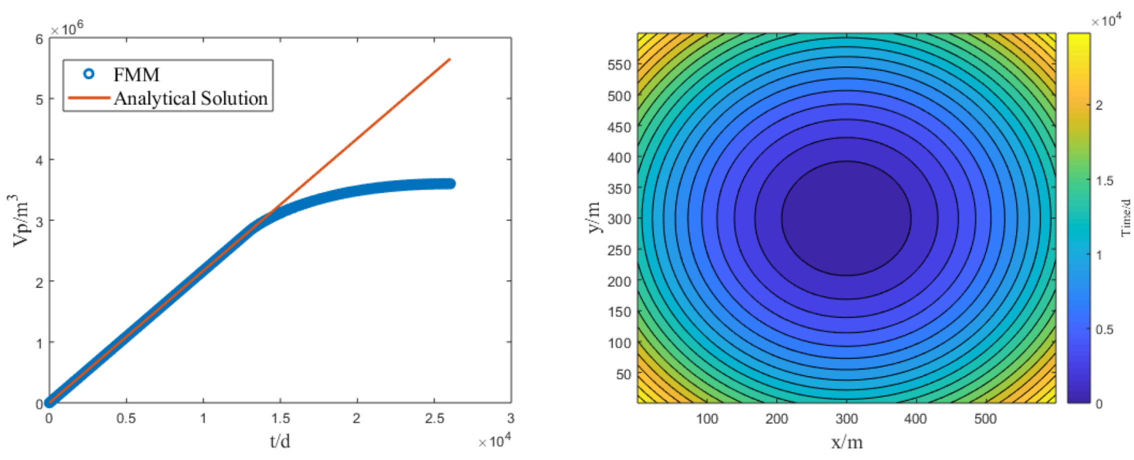
To verify our method, we first apply it to a benchmark case of a vertical well in the center of a square, enclosed reservoir. The results are then compared against the analytical solution for radial flow given by Eq. (15). The key reservoir and fluid properties used in the simulation are listed in Table 1. Figure 1a presents the comparison of the VOI calculated by our method against the analytical solution. During the early-time period ($t < 1.5 \times 10^4$ days), the pressure transient has not yet reached the boundaries, and the system exhibits an infinite-acting radial flow regime. In this phase, our numerical results show excellent agreement with the analytical solution. Conversely, during the late-time period ($t > 1.5 \times 10^4$ days), the pressure front reaches the no-flow boundaries of the model. Consequently, the growth of the VOI calculated by our method decelerates, eventually plateauing as it approaches the total reservoir volume. The analytical solution, being derived for an infinite domain, cannot account for these boundary effects and, as a result, predicts a VOI that continues to increase indefinitely. Furthermore, Figure 1b displays the calculated DTOF map at a specific time. The concentric, circular contours of the DTOF clearly illustrate the propagation of the pressure front, which is perfectly consistent with the expected pattern for radial flow towards the central well.

Table 1—reservoir and fluid properties

reservoir dimensions	600 m × 600 m
permeability (k)	0.01 mD
porosity (ϕ)	0.05
thickness (h)	10 m
oil viscosity (μ)	10 mPa·s
total compressibility (c_t)	1.0×10^{-9} Pa^{-1}

Table 2—reservoir and fluid properties

reservoir dimensions	200x400x30m^3
Matrix permeability (k)	0.001 mD
porosity (ϕ)	0.05
Fracture permeability	10 mD
Fracture half-length	150m
Fracture porosity	0.1
thickness (h)	10 m
oil viscosity (μ)	10 mPa·s
oil compressibility (c_t)	3.0×10^{-9} Pa^{-1}
rock compressibility (c_t)	1.5 × 10-9 Pa^{-1}
Grid dimensions	1 × 1 m^2



(a) (b)

Figure 1—Calculation results for the Volume of Investigation (VOI) in a vertical well exhibiting radial flow. (a) Comparison with analytical solution; (b) Pressure diffusion time.

Comparison with the solution for an infinite conductivity fracture

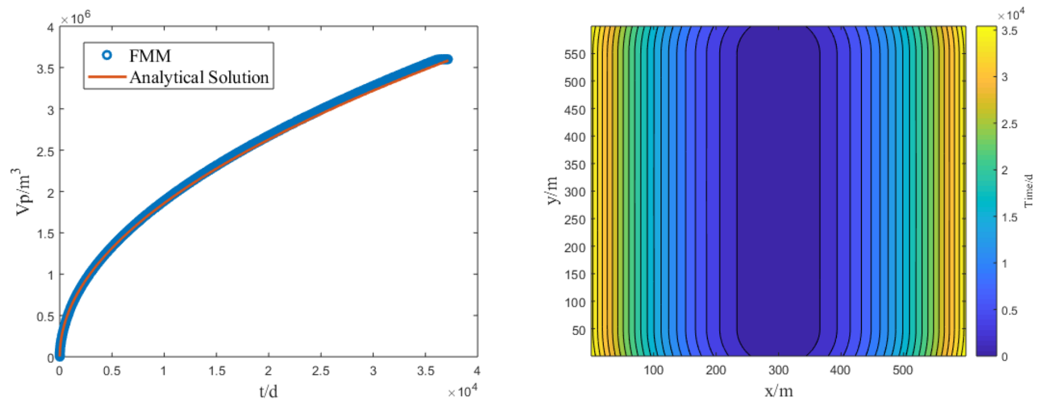
In an infinite-acting reservoir, the flow towards an infinite-conductivity fracture is characterized by a linear flow regime. The corresponding distance of investigation, *r*, is given by:

$$r = \sqrt{2\alpha t} \tag{16}$$

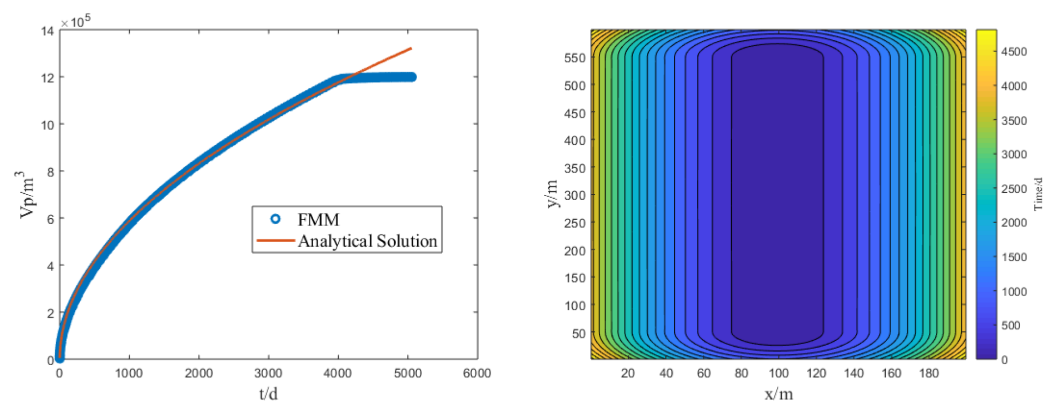
Correspondingly, the Volume of Investigation (VOI), Vp, for this linear flow system is expressed as:

$$V_{\mathrm{p}} = 4L_{\mathrm{f}}rh = 4L_{\mathrm{f}}h\sqrt{2\alpha t} \tag{17}$$

For this verification case, the base reservoir and fluid properties are identical to those used in Section 3.1. We model a single vertical fracture with a half-length (L_f) of 250 m. A high fracture permeability of 1.0 D is assigned to ensure that the infinite-conductivity assumption is valid. To investigate the impact of reservoir boundaries, two scenarios are considered for the reservoir's y-dimension (the dimension parallel to the fracture strike): a large case (y = 600 m) and a small case (y = 200 m).Figure 2a presents the results for the 600 m case. Throughout the entire simulation period, our numerical VOI calculations show an excellent match with the analytical solution for linear flow. In contrast, Figure 2b shows the results for the 200 m case. During the early-time period (t < 4000 days), the system behaves as if it were infinite, and our results align perfectly with the analytical solution, corresponding to the true linear flow regime. However, for the late-time period (t > 4000 days), the pressure transient reaches the boundaries parallel to the fracture. Consequently, the VOI curve calculated by our method deviates from the analytical solution and begins to plateau, correctly approaching the total reservoir volume. This demonstrates our method's capability to accurately capture boundary-dominated flow, a limitation of the classic analytical solution.



(a)



(b)

Figure 2—Comparison of Volume of Investigation (VOI) for a system with an infinite-conductivity fracture under different boundary conditions. (a) Reservoir length along the wellbore axis: 600 m; (b) Reservoir length along the wellbore axis: 200 m.

Sensitivity Analysis

Next, we perform a comprehensive parametric study to investigate the impact of key design and reservoir parameters on the Stimulated reservoir volume (SRV). The analysis focuses on a single hydraulic fracturing stage within a defined reservoir block. The base case parameters for the simulation are detailed in Table 2. In the following sections, we systematically vary the number of fractures, matrix permeability, and fracture conductivity to quantify their influence on the drainage volume.

Characteristics of the change in stimulated reservoir volume (SRV)

Figure 3 illustrates the simulation results for our base case, which features a single hydraulic fracturing stage with four clusters spaced 30 m apart. Figure 3a provides a schematic view of the fracture geometry. Figure 3b plots the Stimulated reservoir volume (SRV) as a function of production time. To facilitate the identification of flow regimes, Figures 3c and 3d present specialized diagnostic plots. Specifically, Figure 3c shows the relationship between SRV and the average Diffusive Time-of-Flight (DTOF), while Figure 3d displays both the time derivative of the SRV and the ratio of this derivative to the SRV itself.A combined analysis of these plots reveals that the evolution of the SRV can be categorized into three distinct stages:

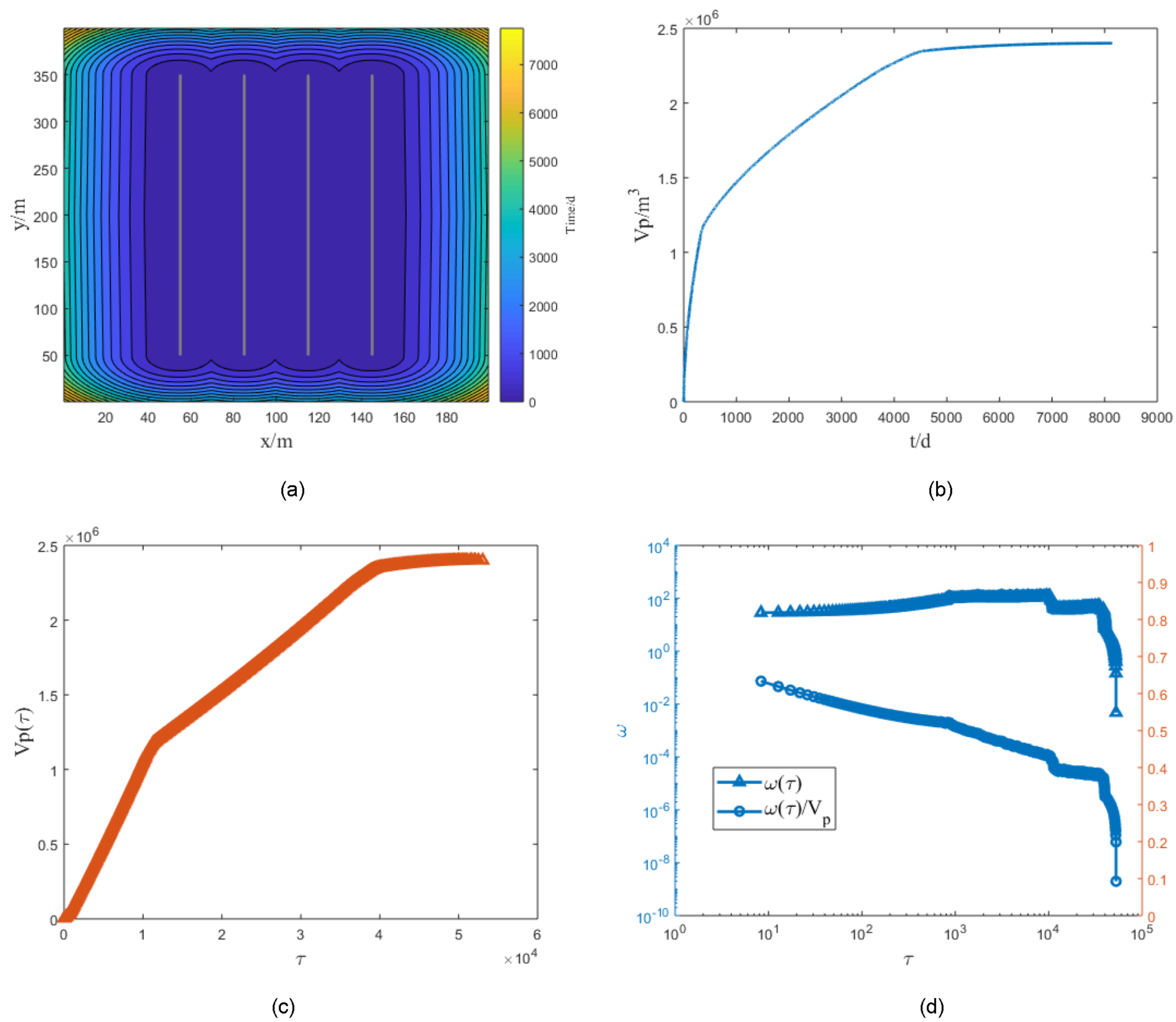


Figure 3—SRV calculation results for a single stage with 4 clusters and 30 m cluster spacing. (a) Diffusive Time-of-Flight (DTOF) distribution; (b) SRV growth as a function of production time; (c) SRV as a function of average DTOF (t); (d) The derivative of Vp ω(t) versus ω(t)/ Vp.

Stage 1: Early-Time Transient Linear Flow (0 – 350 days). In this initial phase, the SRV experiences rapid growth. Each hydraulic fracture independently induces a transient linear flow from the surrounding matrix, and there is no pressure interference between adjacent clusters.

Stage 2: Transitional Flow (350 – 4,600 days). The growth rate of the SRV decelerates as the pressure transients from adjacent fractures begin to interfere and their drainage areas overlap. Despite this interference, the composite flow pattern within the combined stimulated region remains predominantly linear.

Stage 3: Late-Time Boundary-Dominated Flow (t > 4,600 days). The SRV curve effectively reaches a plateau, indicating that the stimulated volume has essentially ceased to expand. The system first transitions to a pseudo-steady state (PSS) flow within the established SRV, and eventually, the entire reservoir response becomes governed by boundary-dominated flow (BDF) as the pressure front interacts with the outer reservoir boundaries.

In unconventional reservoirs, production rates typically decline sharply after 3 to 5 years due to factors such as fracture conductivity degradation and significant reservoir pressure depletion. Consequently, the Stimulated reservoir volume (SRV) achieved within the first 3 years of production serves as a critical and practical metric for evaluating the effectiveness of a hydraulic fracturing treatment. For our base case, the SRV after 3 years (equivalent to approximately 1,100 days) is calculated to be 1.40×10^6 m^3. This volume is then compared against the total Original Oil-In-Place (OOIP) within the single-stage reservoir block, which is 2.40 × 106 m^3 (calculated from a 200m × 400m × 30m block).Based on these values, we define a "drained degree" or "volumetric sweep efficiency" as the ratio of the 3-year SRV to the total OOIP. For this case, the drained degree is 58.3% (1.40×10^6 / 2.40×10^6). This result indicates that nearly half of the reservoir volume within the stage remains unstimulated and is not effectively contributing to production. This observation highlights a clear need for optimization, such as by increasing the number of clusters and reducing the cluster spacing, to improve the overall stimulation effectiveness.

Impact of Cluster Number(or Cluster Spacing)

In this section, we systematically investigate the impact of fracture density on the development of the Stimulated reservoir volume (SRV). To achieve this, we compare the performance of four distinct completion designs, which are defined as follows:Case 1 (Base Case): 4 clusters with 30 m spacingCase 2: 6 clusters with 30 m spacingCase 3: 10 clusters with 20 m spacingCase 4 (High-Density Case): 20 clusters with 10 m spacing

Figure 4 presents the simulation results, where the number of clusters is increased to 6 while maintaining the 30 m spacing. For this completion design, the SRV after 3 years of production increases to 2.00×10^6 m^3. This corresponds to an drained degree of 83.3%, a significant improvement over the base case. The diagnostic plot in Figure 4b confirms that the SRV evolution still follows the three-stage pattern. However, a direct comparison with the base case (Figure 3b) reveals two key changes: the duration of the intermediate transitional flow stage is significantly reduced, and consequently, the final stage of pseudo-steady state (PSS) flow begins much earlier. This observation demonstrates that increasing the fracture density accelerates the onset of inter-fracture interference, causing the system to reach the stabilized, late-time flow regime more quickly.

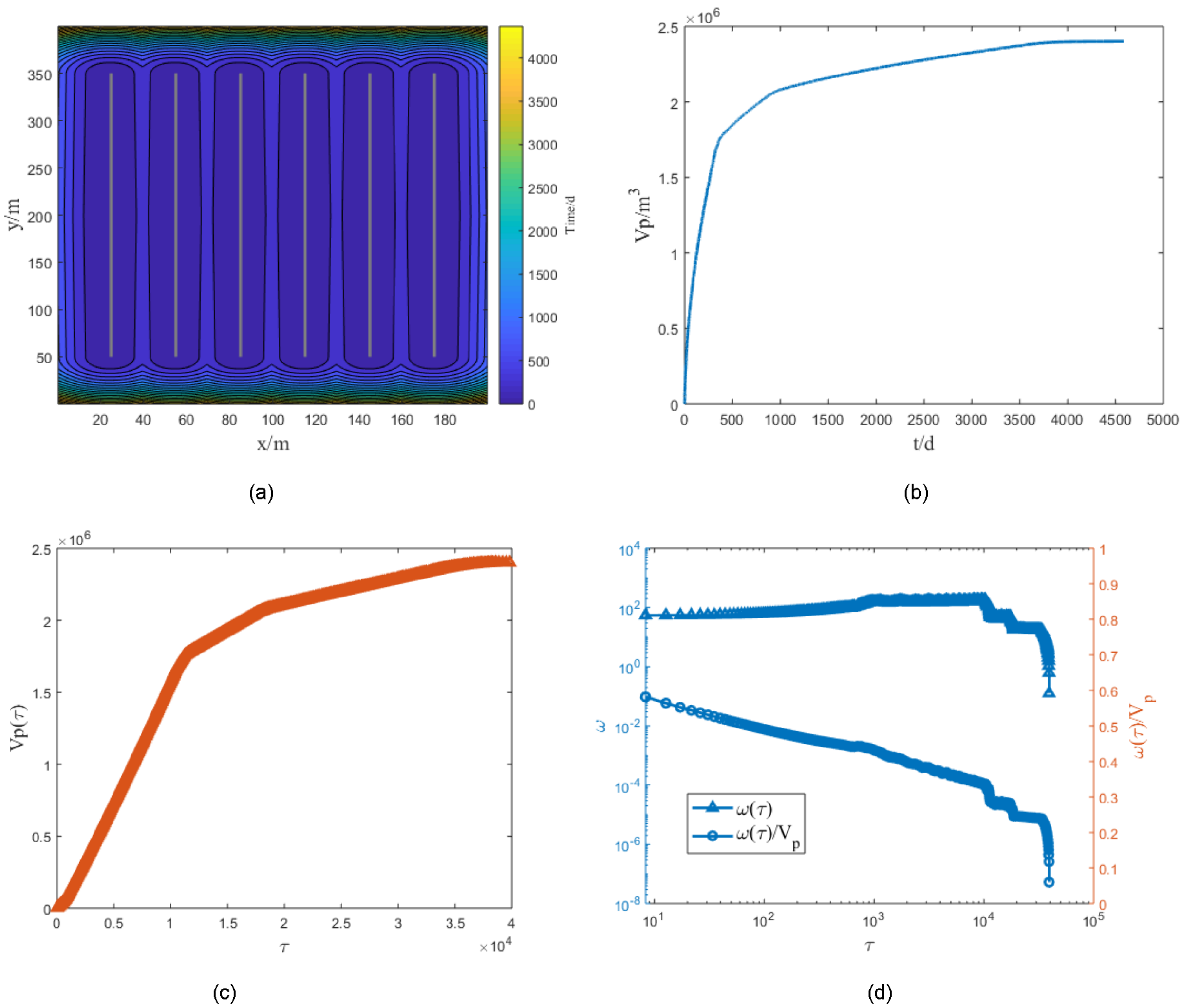


Figure 4—SRV calculation results for a single stage with 6 clusters and 30 m cluster spacing. (a) Diffusive Time-of-Flight; (DTOF) distribution (b) SRV growth as a function of production time; (c) SRV as a function of average DTOF (t); (d) The derivative of Vp ω(t) versus ω(t)/ Vp ω(t)/Vp.

Figure 5 compares the SRV growth curves for the four completion designs, illustrating the direct relationship between cluster and the SRV. As the number of clusters increases (i.e., fracture density increases), the initial rate of SRV growth becomes significantly steeper, leading to an earlier onset of inter-fracture interference. The plot shows that after only 500 days of production, the SRV for the high-density cases (10 and 20 clusters) already begins to plateau. By day 1,000, the SRV curves for the 6-, 10-, and 20-cluster designs have largely converged, indicating they are all approaching a similar ultimate stimulated volume. It is important to note that these simulations are based on a matrix permeability of 0.001 mD. The influence of permeability, particularly in ultra-low permeability systems like shales (micro- to nano-Darcy range), will be specifically addressed in Section 4. A key insight from this analysis is that for high-density designs (i.e., the 10- and 20-cluster cases), the SRV is already approaching its maximum possible value by the time significant inter-fracture interference begins. This suggests a powerful simplification for modeling: for such intensively fractured reservoirs, the ultimate stimulation effectiveness can be reasonably estimated by analyzing the drainage performance of a single, isolated fracture element, as the system rapidly behaves like one large, composite stimulated region.

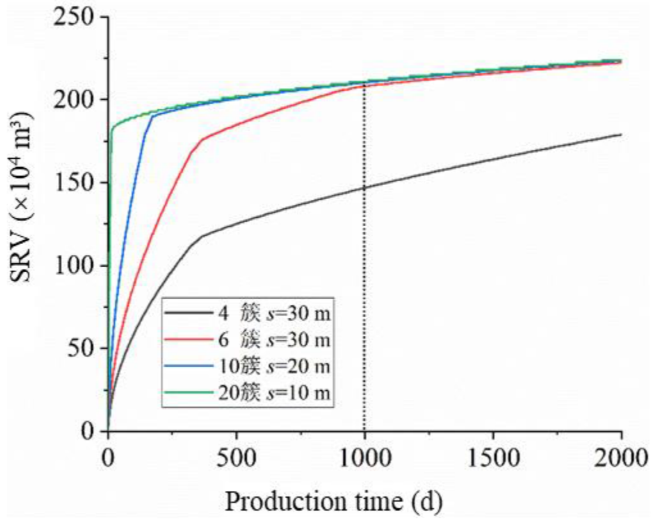


Figure 5—Impact of cluster number on the evolution of stimulation effectiveness.

Formation permeability

In this section, we investigate the influence of matrix permeability (km) on SRV, a critical factor for production in unconventional reservoirs. Four different matrix permeability values, spanning three orders of magnitude to cover the range from conventional low-permeability to shale-like nano-Darcy systems, are simulated: (1) k_m=10^3 mD; (2) k_m=10^{-4} mD; (3) k_m=10^{-5} mD; (4) k_m=10^{-6} mD. For this sensitivity study, the high-density completion design (20 clusters with 10 m spacing) identified in the previous section is used as the base fracture geometry. All other reservoir and fluid properties remain consistent with the base model described in Section 4.1.

Figure 6 displays the SRV distribution maps after 3 years of production for the four permeability cases, using the high-density (10 m spacing) fracture design. In these maps, the blue regions represent the portion of the reservoir effectively stimulated and drained, while the yellow areas correspond to the undrained volume that remains unaffected by pressure depletion. A direct comparison across the four scenarios reveals a clear and critical trend: stimulation effectiveness is highly sensitive to matrix permeability, with lower permeability resulting in a drastically reduced SRV. This effect becomes particularly pronounced in the ultra-low permeability cases. For matrix permeabilities of 10^{-5} mD and 10^{-6} mD (nano-Darcy scale), the pressure drawdown is confined to a very narrow zone, extending only about 3 meters from each fracture face. This observation is a key finding of our study. It demonstrates that even with an aggressive 10-meter cluster spacing, the individual stimulated regions around each fracture fail to connect or interfere. Consequently, a substantial volume of the reservoir between the fractures remains completely unstimulated, indicating that this high-density design is still insufficient to achieve effective volumetric sweep in such tight, nano-Darcy systems.

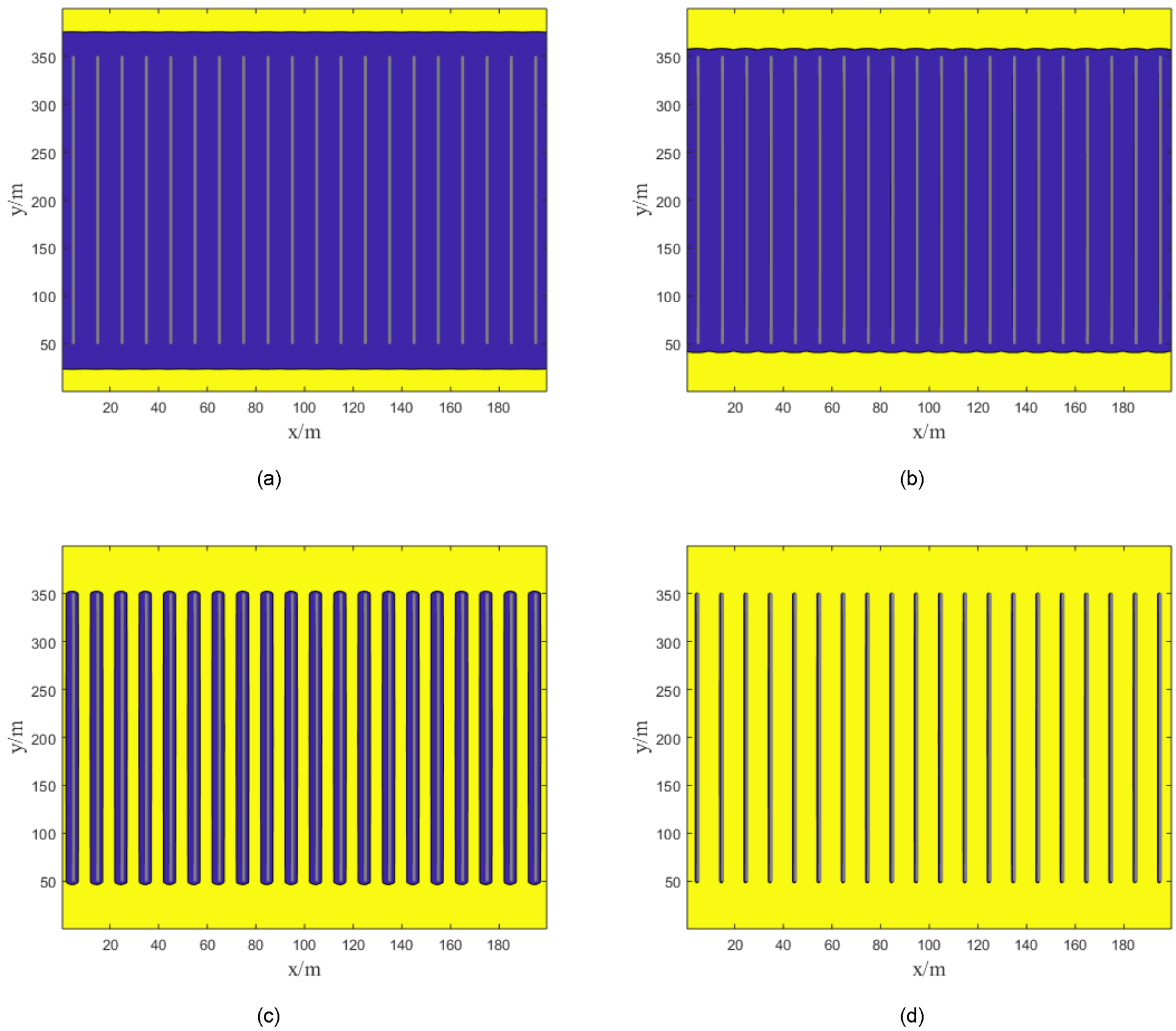


Figure 6—Impact of matrix permeability on SRV (Blue regions represent the drained volume, while yellow indicates the undrained reservoir). (a) k_m=10^3 mD; (b) k_m=10^{-4} mD; (c) k_m=10^5 mD; (d) k_m=10^6 mD.

Figure 7 quantifies the impact of matrix permeability on the SRV. The plot clearly shows that as permeability decreases, both the speed of SRV growth and the final achievable SRV are severely curtailed. After 3 years of production (approximately 1,100 days), the SRV of the four permeability formations are 210.6 × 10^4 m3, 191.4 × 10^4 m3, 94.2 × 10^4 m3, and 3.7 × 10^4 m3 (from high to low), corresponding to 87.8%, 80.0%, 39.3%, and 1.5% of volumetric sweep efficiency, respectively.

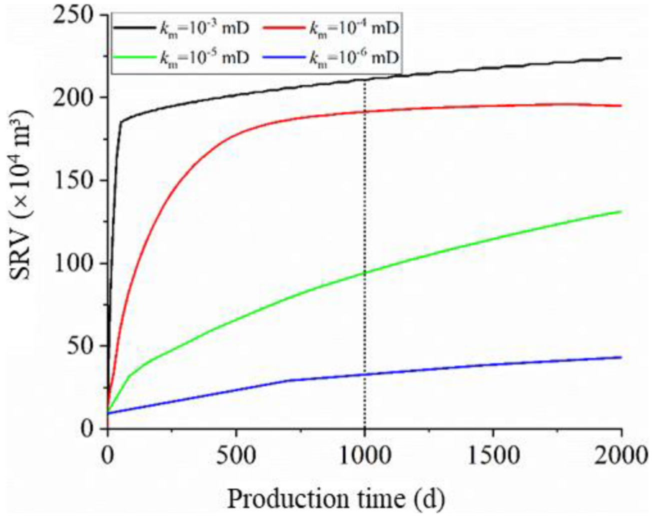


Figure 7—Impact of matrix permeability on the evolution of Stimulated reservoir volume (SRV) over time.

Fracture conductivity

Dimensionless fracture conductivity (C_D), a critical parameter that governs flow regimes and well productivity, is the focus of this section. Our primary objective is to investigate how the requirement for fracture conductivity changes in the context of modern, high-density fracture completions. To achieve this, the analysis is twofold: we first evaluate the impact of C_D on the Stimulated reservoir volume (SRV) for both a single-fracture scenario and a multiple-fracture scenario (representing intensive fracturing). Then, by comparing the results of these two scenarios, we aim to reveal the trend of conductivity demand as fracture density becomes higher.

Single-Cluster. The analysis begins with the fundamental case of a single, isolated fracture. We simulate the SRV for this single fracture under four distinct levels of C_D: 1, 5, 15, and 20.In the model, these varying C_D values are achieved by systematically adjusting the fracture permeability, while all other reservoir and completion parameters are held constant as defined in the base case.

Figure 8 visually demonstrates the profound impact of dimensionless fracture conductivity (C_D) on the shape and extent of the Stimulated reservoir volume (SRV) for a single fracture. At low conductivity levels (e.g., C_D = 1, Figure 8a), the fracture itself acts as a significant bottleneck to flow. This high pressure drop along the fracture chokes production from its distal parts (the tips), causing the SRV to be confined to a diamond-shaped region concentrated around the wellbore. As C_D increases, the fracture becomes more efficient at transporting fluids, allowing for a more uniform drainage pattern along its entire length. The SRV expands, transitioning to a wider, elliptical geometry (Figure 8b). At a C_D of approximately 15 (Figure 8c), the system reaches what is effectively the infinite-conductivity limit. Beyond this critical threshold, any further increase in conductivity (e.g., to C_D = 20, Figure 8d) yields only marginal gains in the stimulated volume, as the primary constraint on flow shifts from the fracture to the surrounding matrix.

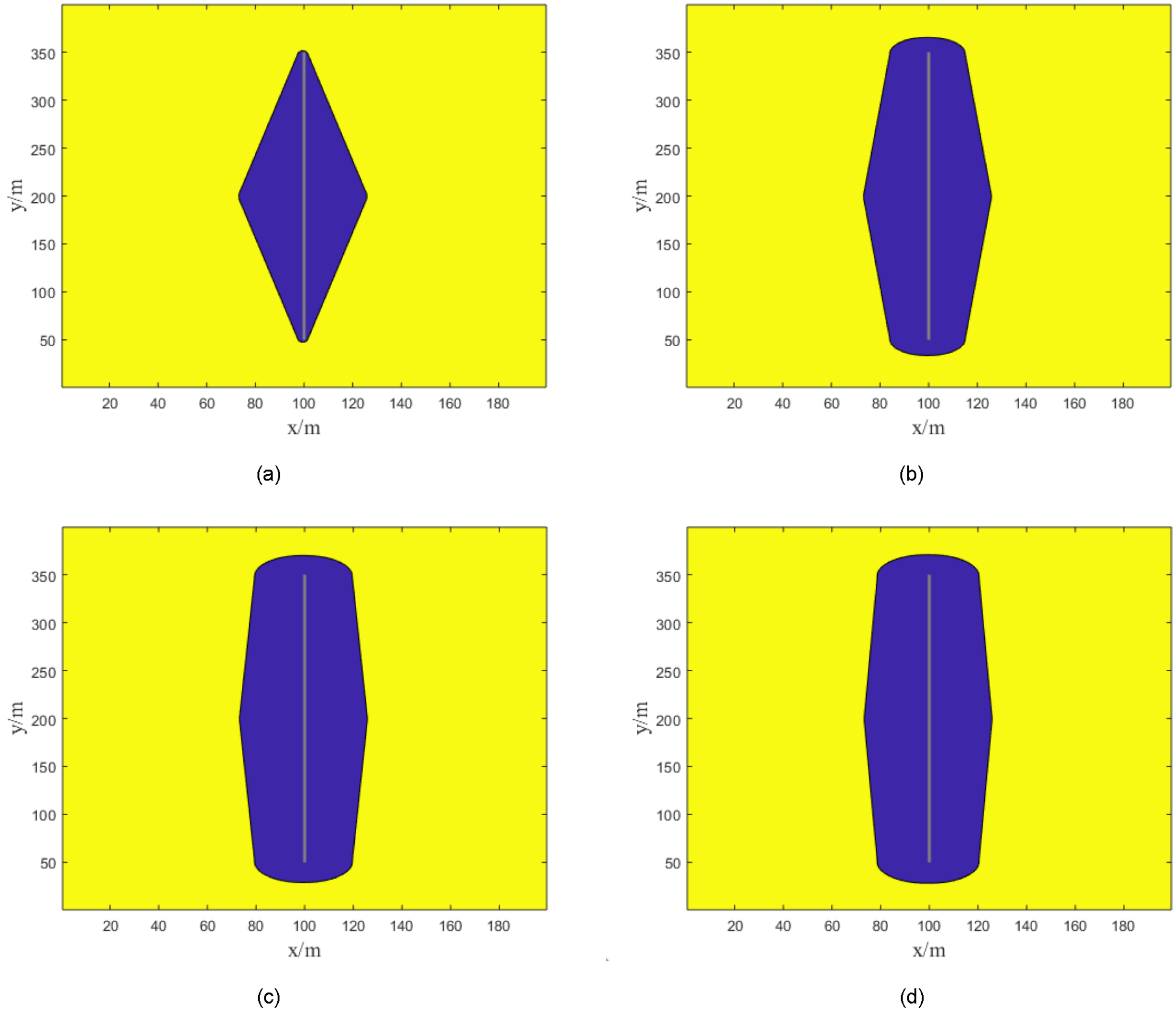


Figure 8—Impact of dimensionless fracture conductivity on SRV (Single-Fracture). (a) CD=1; (b) CD=5; (c) C_D=15; (d) C_D=20

Multiple clusters (6_s 10_s 20 clusters). Building upon the single-fracture analysis, this section extends the investigation to the more complex and practical multiple-fracture scenarios. We examine the impact of dimensionless conductivity (C_D) for the three established completion designs: 6, 10, and 20 clusters, which correspond to fracture spacings of 30 m, 20 m, and 10 m, respectively. Figures 9, 10, and 11 show the SRV maps for the 6-, 10-, and 20-cluster cases with different conductivities. Figure 12 shows the SRV vs. production time curves. Table 1 lists the calculated SRV results. Figure 13 analyzes the sensitivity of increasing conductivity under different spacings, using the ratio of the SRV to the maximum SRV at threshold conductivity (Vp/Vpmax). In this analysis, a curve that flattens out more quickly (i.e., exhibits a smaller rate of change) signifies that increasing fracture conductivity beyond a certain point yields diminishing returns for that particular fracture design.

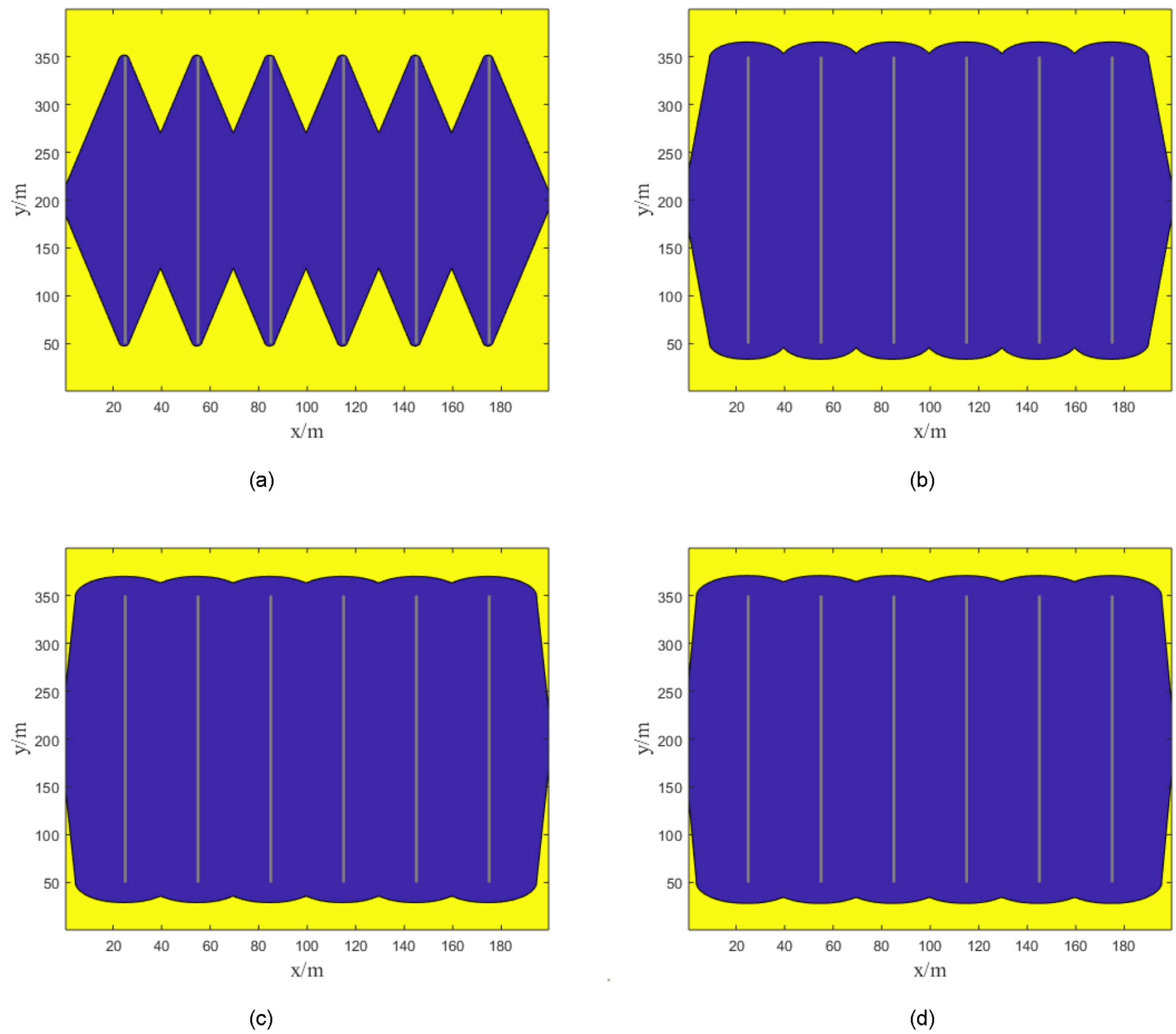


Figure 9—Effect of dimensionless fracture conductivity (C_D) on SRV distribution for the 6-cluster (s = 30 m) case. (a) C_D =1; (b) C_D =5; (c) C_D =15; (d) C_D =20.

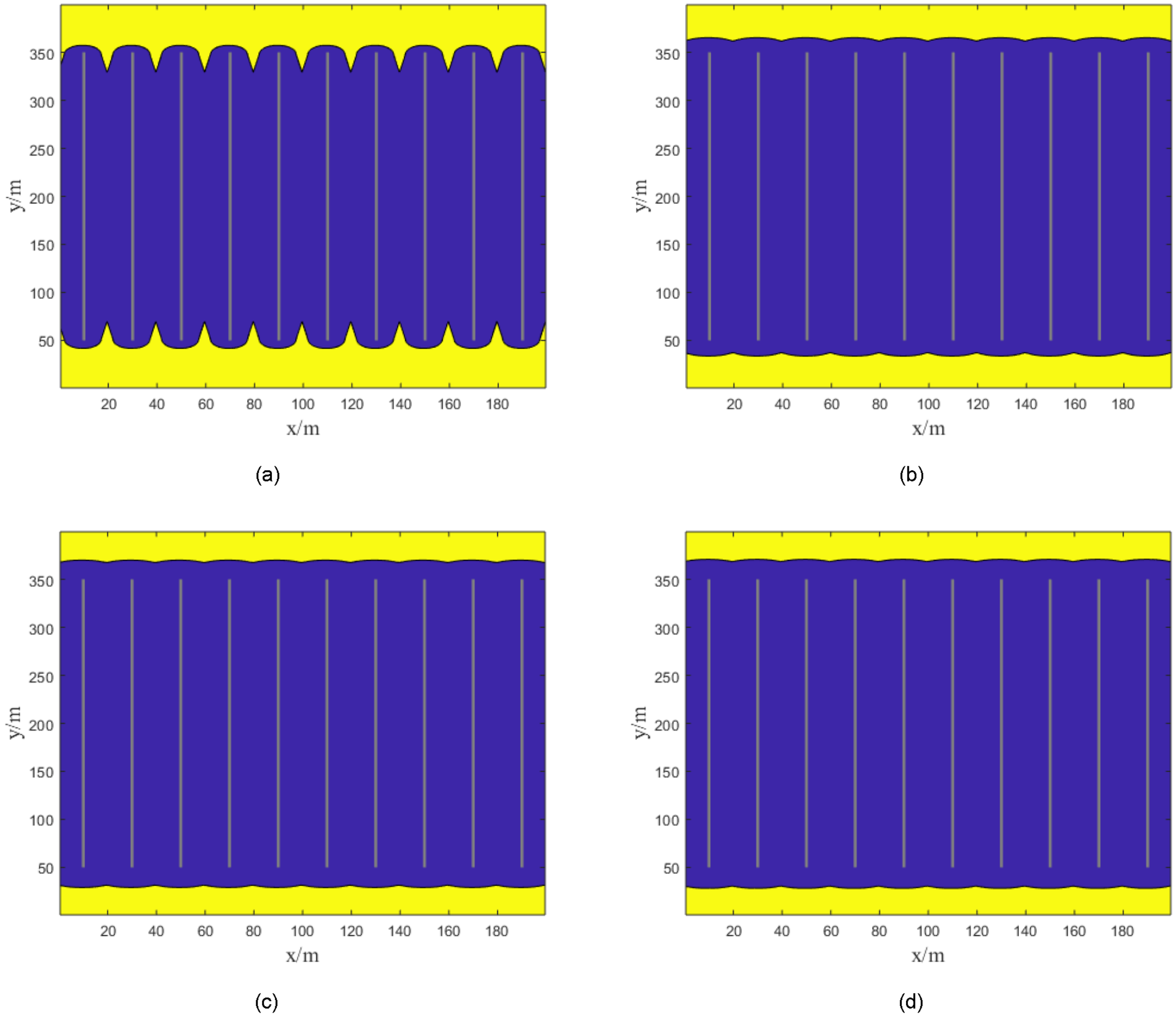
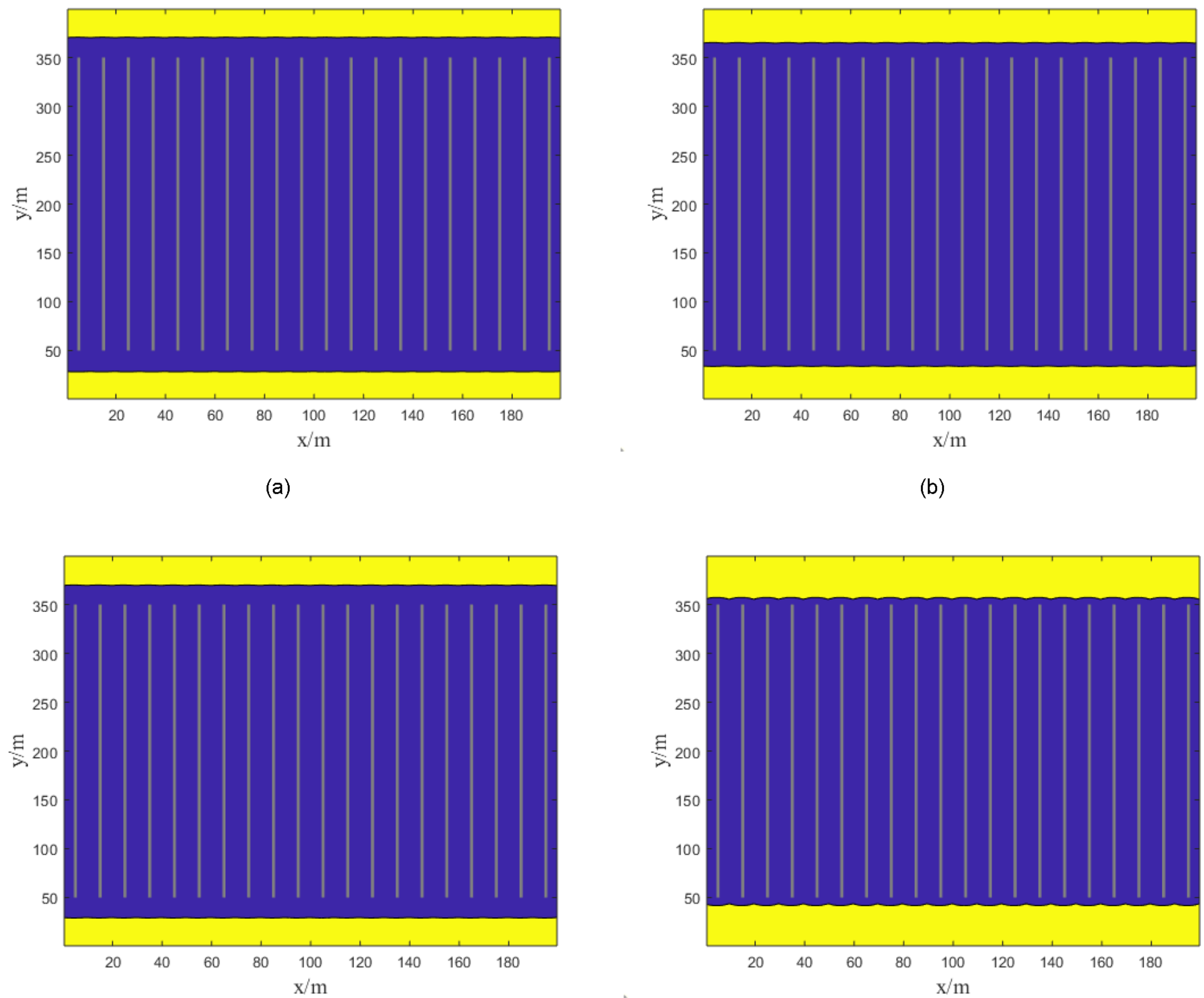


Figure 10—Effect of dimensionless fracture conductivity (C_D) on SRV distribution for the 10-cluster (s = 20 m) case. (a) C_D =1; (b) C_D =5; (c) C_D =15; (d) CD =20.



(a) (b)

(c) (d)

Figure 11—Effect of dimensionless fracture conductivity (C_D) on SRV distribution for the 20-cluster (s = 10 m) case. (a) C_D =1; (b) C_D =5; (c) C_D =15; (d) C_D =20.

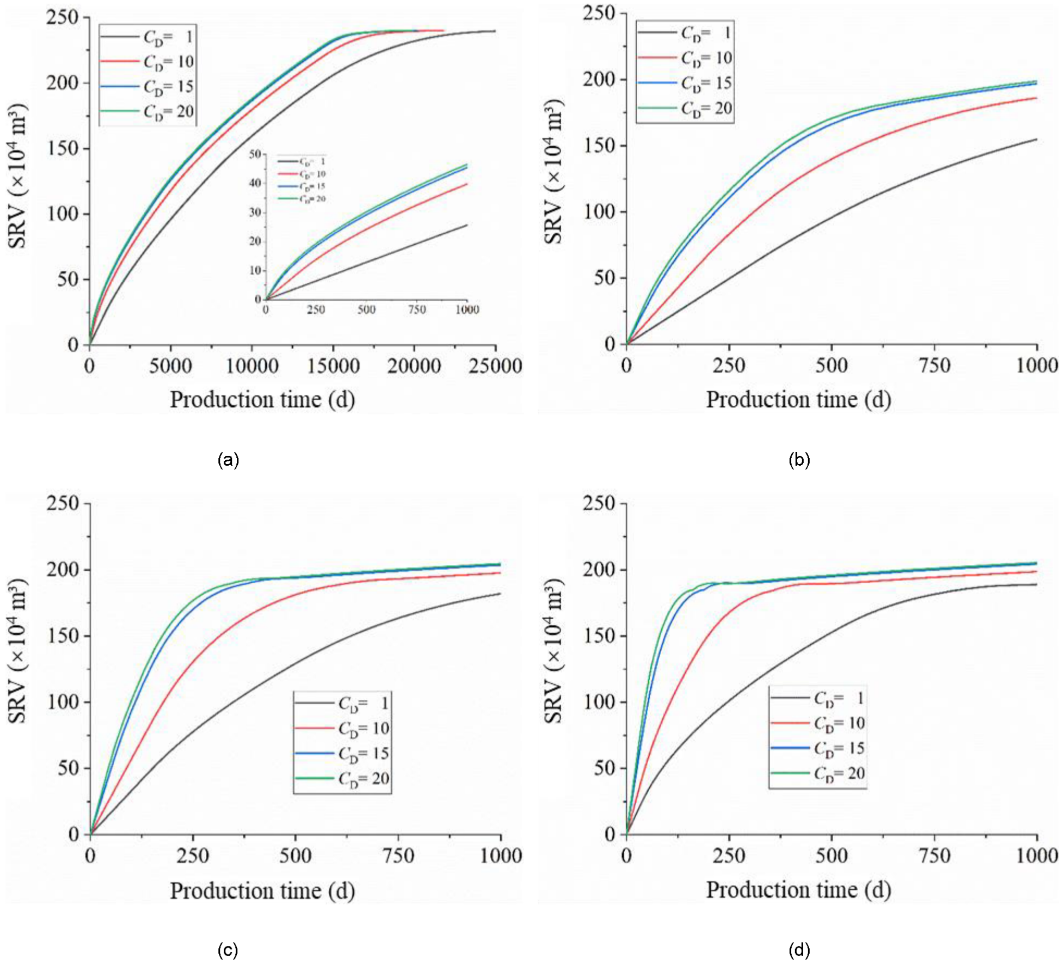


Figure 12—Comparison of SRV evolution over time for different C_D values across all fracture spacing scenarios. (a) single cluster; (b) 6 clusters; (c) 10 clusters; (d) 20 clusters.

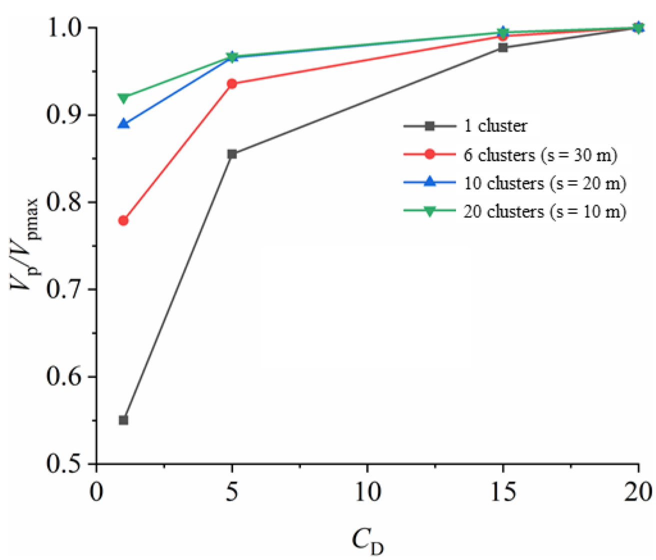


Figure 13—Normalized analysis of conductivity sensitivity: plotting the SRV performance ratio (V_p/V_{pmax}) against C_D for each fracture spacing case.

Figure 9 presents the SRV maps for the 6-cluster (s=30 m) completion design. With a low dimensionless conductivity of C_d = 1 (Figure 9a), the pressure drop along the fractures is substantial. As a result, the individual drainage areas around each fracture fail to effectively coalesce, leaving significant undrained zones between the fractures, with a corresponding stimulation efficiency of only 64.52%. By increasing the conductivity to C_D= 5 (Figure 9b),the stimulated zones successfully merge, dramatically boosting the stimulation efficiency to 77.51%. However, further increases in conductivity yield diminishing returns. As C_D is increased to 15 and then 20, the stimulation efficiency essentially plateaus at 82.03% and 82.83%, respectively. (The specific calculated values for all cases are detailed in Table 3).

Table 3—Summary of final Stimulated reservoir volume (SRV) and Stimulation Efficiency for all simulated C_D and fracture spacing cases

C_D	SRV (10^4 m^3)				drained degree (%)				V_p/V_{pmax}			
	single cluster	6 clusters	10 clusters	20 clusters	single cluster	6 clusters	10 clusters	20 clusters	single cluster	6 clusters	10 clusters	20 clusters
1	25.65	154.85	181.95	188.98	10.69	64.52	75.81	78.74	0.55	0.78	0.89	0.92
5	39.87	186.02	197.62	198.55	16.61	77.51	82.34	82.73	0.86	0.94	0.97	0.97
15	45.56	196.88	203.52	204.26	18.98	82.03	84.80	85.11	0.98	0.99	0.99	0.99
20	46.62	198.78	204.62	205.34	19.43	82.83	85.26	85.56	1.00	1.00	1.00	1.00

Figure 10 displays the SRV distribution for the 10-cluster (s=20 m) design. A key trend emerges when comparing this scenario to the wider 30 m spacing case (Figure 9): as fracture density increases, the demand for ultra-high fracture conductivity diminishes. Even at a low conductivity of C_D = 1, the performance is markedly improved. Because the fractures are closer together, the undrained zones between them are significantly reduced. The stimulation efficiency at this stage is 75.81%, a substantial improvement over the 64.52% seen in the 30 m spacing case. The stimulation efficiency begins to plateau around C_D = 10, reaching 82.34%. Further increasing the conductivity to C_D = 20 only provides a marginal gain, with the efficiency rising to 85.26%.

Figure 11 illustrates the SRV results for the highest density design: 20 clusters with a tight 10-meter spacing. A striking observation from the SRV maps is that fracture conductivity has become a negligible factor in this high-density scenario. The four cases, from C_D = 1 to C_D= 20, produce nearly identical SRV distributions. The increase in the maximum achievable stimulation efficiency is marginal, with a mere 0.3% gain over the 10-cluster case. This leads to a crucial conclusion for this 0.001 mD reservoir: after a 3-year production period, there is virtually no difference in the stimulation effectiveness between the 10-cluster (20 m spacing) and the 20-cluster (10 m spacing) designs. This strongly indicates that the system has reached a point of diminishing returns.

Figure 12 compiles the SRV growth curves for all simulated scenarios, providing a comprehensive overview of the interference between fracture conductivity and fracture spacing. A clear visual trend emerges from these plots: as fracture spacing decreases (i.e., as fracture density increases from a single fracture to 20 clusters), the divergence between the performance curves for different C_D values diminishes significantly. In the 20-cluster case (Figure 12d), the curves for C_D = 10, 15, and 20 are nearly indistinguishable. To quantify this crucial observation more rigorously, a normalized performance metric is presented in Figure 13. For each fracture spacing design, we define a performance ratio, Vp/V_{pmax}. Here, V_p is the stimulated volume at a given C_D, and V_{pmax} is the maximum potential SRV, represented by the value achieved with a C_D of 20 (approximating the infinite-conductivity case). A narrow range for this ratio indicates that even lowconductivity fractures perform close to the maximum potential, making high conductivity less critical. The results of this normalization, derived from Figure 13, are as follows: 0.56~1 for single fracture (which can be considered as infinite fracture spacing), 0.78~1 for 30m fracture spacing, 0.89~1 for 20m fracture spacing, and 0.92~1 for 10m fracture spacing. This quantitative analysis provides compelling evidence for the final, critical conclusion: intensive fracturing (i.e., high-density, closely spaced fractures) significantly lowers the required fracture conductivity for effective reservoir stimulation.

Field Case Application

To validate the reliability of the numerical model and its core finding—that high fracture density enhances stimulation effectiveness— a comparative field pilot was conducted in the Chang 7 shale oil play in the Ordos Basin. The reservoir in this area is characterized by typical ultra-low permeability and nano-scale pore throats, which closely aligns with the low- to ultra-low permeability scenarios analyzed in our numerical simulations.

The study involved two adjacent horizontal well pads, H85 and H86, which share similar reservoir properties and fluid characteristics, ensuring a reliable comparison. The Test Group, on the H85 pad (4 wells), was completed using a high-density design optimized with our model, featuring a cluster spacing of 6 m. The Control Group, on the H86 pad (4 wells), utilized the conventional fracture design for the area with a cluster spacing of 12 m. All horizontal wells on both pads have a lateral length of 1,500 m. Production data (Fig. 14) reveals a stark contrast in performance. After two years of production, the average cumulative oil production per well on the H85 pad reached 8,785.5 t, representing a 31.3% increase compared to the H86 pad. This field result empirically demonstrates that tighter cluster spacing promotes the development of a more uniform and extensive fracture network, leading to a significant enhancement in well productivity. This successful application not only corroborates the accuracy and efficiency of the proposed FMM-based calculation method but also highlights its practical value as a powerful tool for optimizing fracture designs in field development.

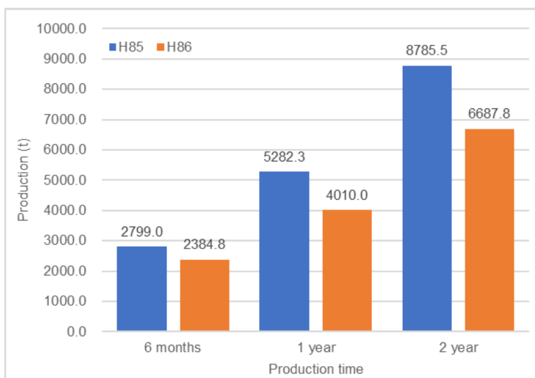


Figure 14—Production history of H85&H86 pad.

Conclusions

In this study, the Fast Marching Method (FMM) was employed to solve the pressure front diffusion equation, enabling the dynamic tracking of the Stimulated reservoir volume (SRV) over time and the subsequent calculation of the stimulation efficiency. The accuracy of this methodology was rigorously validated against analytical solutions. Based on this robust numerical framework, a systematic analysis of key fracture and formation parameters was conducted, yielding the following principal conclusions:

1. The Fast Marching Method (FMM) is demonstrated to be a computationally efficient and powerful tool for the optimization of hydraulic fracture design parameters. Its high speed makes it ideal for running the numerous simulations required for sensitivity analysis and field development planning.
2. Diagnostic plots, such as the SRV curve and its corresponding time derivative, can serve as effective tools for identifying the dominant flow regimes (e.g., linear flow, boundary-dominated flow) within the reservoir during different production periods.

3. Formation permeability is the most critical parameter governing the ultimate stimulation efficiency. In ultra-low permeability reservoirs (e.g., micro- to nano-Darcy), where the pressure diffusion distance is extremely limited (calculated to be less than 3 meters from the fracture face after 3 years for the parameters studied), aggressive reduction of fracture spacing is the primary and most effective strategy for successful development.
4. Creating a high-density fracture network significantly lowers the requirement for high fracture conductivity. As fractures are placed closer together, the system becomes less sensitive to the conductivity of individual fractures, allowing for effective stimulation even with lower dimensionless conductivity values.

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